

Database Design Document

Student Course Allocation System

1. Overview

The Student Course Allocation System uses PostgreSQL as the relational database to store administrator accounts, student records, course information, and course allocation results. The database is designed to maintain data integrity, minimize redundancy, and support efficient querying for the allocation process.

2. Database Tables

2.1 Admins

Purpose:

Stores administrator login credentials used for system authentication.

Column	Data Type	Description
id	SERIAL	Primary Key
username	VARCHAR(100)	Unique administrator username
password_hash	VARCHAR(255)	Encrypted password
created_at	TIMESTAMP	Record creation time
updated_at	TIMESTAMP	Last update time

2.2 Courses

Purpose:

Stores available courses and seat distribution.

Column	Data Type	Description
id	SERIAL	Primary Key
course_code	VARCHAR(20)	Unique course code
course_name	VARCHAR(255)	Course name
description	TEXT	Course description
total_seats	INT	Total seats available
general_seats	INT	General category seats
obc_seats	INT	OBC category seats

Column	Data Type	Description
sc_seats	INT	SC category seats
st_seats	INT	ST category seats
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

2.3 Students

Purpose:

Stores student information, marks, category, application date, and preferred courses.

Column	Data Type	Description
id	SERIAL	Primary Key
student_id	VARCHAR(20)	Unique student ID
name	VARCHAR(255)	Student name
marks	DECIMAL(5,2)	Student marks
category	ENUM	Reservation category
application_date	TIMESTAMP	Application submission date
pref1_course_id	INT	First course preference
pref2_course_id	INT	Second course preference
pref3_course_id	INT	Third course preference
created_at	TIMESTAMP	Created time
updated_at	TIMESTAMP	Updated time

2.4 Allocations

Purpose:

Stores the final course allocated to each student.

Column	Data Type	Description
id	SERIAL	Primary Key
student_id	INT	References Students table

Column	Data Type	Description
course_id	INT	References Courses table
preference_rank	INT	Preference (1, 2 or 3)
category_slot	ENUM	Allocated category
status	ENUM	Allocation status
allocated_at	TIMESTAMP	Allocation time